Partner-Specific Affective Precision in Social Active Inference

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Abstract. In multi-agent social settings, model reliability varies across relationships. Beyond inferring what others will do, an agent must calibrate how confidently those inferences should guide policy selection for each relationship. An agent may maintain a well-validated model of one partner, a fragile model of another, and a model under revision for a third; collapsing these into a single confidence estimate loses information critical for policy selection. Thus, we use active inference to model affect as a metacognitive estimate of confidence in the current partner model, formalized as relationship-specific affective precision. Each partner's behavioral evidence updates a local confidence estimate that modulates policy precision during selection, regulating how strongly inferred beliefs are expressed in policy rather than changing the beliefs themselves.
Simulations in a multi-partner graded trust game show that partner-local affective precision influences behavior through policy commitment rather than improved inference about partner states. Because the mechanism tracks partner-response predictability rather than realized payoff, greater confidence leads to more committed policies without necessarily producing higher rewards. Under abrupt shifts in social behavior, confidence accumulated from a previously reliable relationship can continue to sharpen commitment after the partner changes, showing that adaptation requires revising confidence estimates and reallocating engagement across relationships. Finally, varying precision gain and priors on model fitness reveal distinct trust-calibration profiles, generating computational hypotheses about individual differences in social trust. Together, these results demonstrate that affective precision serves as a relationship-specific metacognitive signal that distinguishes social prediction from social commitment.

Keywords: Active inference · Affective precision · Social metacognition · Predictive reliability · Policy precision · Trust calibration

1 Introduction

Social agents rarely act on a single, stable model of the social world; instead, they maintain different expectations for different partners. One partner may be familiar and reliable, another newly encountered and uncertain, and another predictable but adversarial. Repeated social exchange depends on learning such partner-specific regularities, including reciprocity, reputation, moral character, and social value [16, 3, 7]. In such settings, the question is not only what an agent believes about each partner, but how confidently those beliefs should guide policy inference.

Computational models of social decision-making can therefore be separated into three problems. The first is inferring a partner's behavioral disposition: whether their actions are cooperative, exploitative, reciprocal, or volatile. The second is modeling what they know or intend: the theory-of-mind problem of how another agent represents the world and plans action [10, 11, 20]. The third is estimating whether the resulting partner model is reliable enough to guide confident policy commitment [22]. To address this third problem, an agent must decide not only what it believes about a partner, but how strongly to act on those beliefs. However, many formal models specify the content of social inference while treating confidence in those inferences as fixed, global, or implicit.

The active inference framework [18] provides a natural formalism for this problem because it links perception, belief updating, and policy selection through a common inferential process. In active inference, agents infer hidden states of the world and select policies by minimizing expected free energy. Moreover, policy inference is precision-weighted, controlling how sharply an agent commits to its inferred policy distribution, making it an effective model of not only what an agent believes but how confidently those beliefs are expressed in policy.

Existing social active-inference models address complementary parts of the problem. Theory-of-mind models address how agents model what others know or intend through recursive belief structures and factorized generative models [20, 21]. Volatility and social-learning models address how prediction errors should change belief updating under uncertainty [17, 3, 8]. Social-learning work also shows that agents revise expectations from discrepancies between expected and observed behavior of others, including in trust-game interactions [15]. These approaches clarify how agents infer partners, model others' mental states, and revise beliefs under uncertainty. However, they do not by themselves explain how confidence in a social model should remain local to the relationship that generated the evidence.

Affect provides a natural candidate mechanism for this confidence-calibration role. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations and social commitments [2]. Free-energy and active-inference accounts extend this view by linking affective dynamics to prediction error, model evidence, emotional-state inference, and precision-weighted policy inference [14, 24, 13, 19]. Prior affective-precision accounts make confidence endogenous by relating affect to subjective model fitness and policy precision [13]. On this view, affect can regulate how strongly current beliefs are expressed in policy, shaping the degree of commitment rather than the content of inference.

We therefore define partner-local affective precision as a relationship-specific confidence signal. The focal agent maintains a separate β_k estimate for each partner, updated only from that partner's behavioral evidence, which modulates policy precision when inferring policies toward that partner. Unlike prior affective-precision accounts centered on policy-inference quantities [13], we formulate the affective signal from partner-response likelihood surprisal: the surprise of an observed partner response under the focal agent's pre-update predictive posterior over that partner's type and stance. This makes the signal perceptual and relationship-local: it tracks whether the current model predicts this partner's behavior, preserving predictability differences that a global confidence estimate would collapse.

We evaluate this mechanism in a repeated multi-partner graded trust game with partner-choice conditions, abrupt betrayal, and profile variation. The remainder of this paper is organized as follows: Section 2 specifies the multi-partner trust game, per-partner generative models, and the partner-local affective precision mechanism. Section 3 reports simulation results and analyses. Section 4 discusses implications, limitations, and directions for future work.

2 Methods

A focal active-inference agent plays a repeated multi-partner trust/investment game with four partners whose behavioral types and stances are hidden. The agent is implemented using the `pymdp` library for discrete active inference [12, 6], which provides standard components for belief updating and policy selection. Partner-local affective precision is layered on top of separate per-partner generative models, allowing both evidence and policy precision to remain relationship-specific. The task generative process, focal-agent POMDP, affective-precision update, and simulation protocols are specified in Appendices A–D.

2.1 Trust game

We evaluate the mechanism in a repeated multi-partner trust/investment game [4]. The task has a Prisoner's-Dilemma-style social-dilemma structure: investing more can yield larger returns when a partner cooperates, but larger losses relative to the risk-free baseline when the partner defects. The task therefore requires the focal agent to solve two coupled problems: predicting how each partner is likely to respond, and deciding how strongly to commit to that prediction through investment.

Each interaction consists of partner assignment or selection, investment, and response. In partner-choice conditions, the focal agent first selects a partner k; in conditions without explicit partner choice, the interaction partner is assigned according to the configured protocol. The agent then chooses a graded investment level $a \in \mathcal{A} = \{0, 1, 2, 3, 4, 5\}$, where 0 denotes no investment and 5 denotes maximum investment. The selected partner emits a response $o_k^{\text{act}} \in \{\text{cooperate}, \text{defect}\}$, sampled from that partner's hidden behavioral state.

Each partner has two hidden state factors. The first is behavioral type, $s_k^{\text{type}} \in \{\text{cooperator}, \text{reciprocator}, \text{exploiter}, \text{random}\}$. These labels are stylized abstractions of heterogeneous motives and reciprocity patterns observed in trust-game behavior [9, 23]. The second is stance toward the focal agent, $s_k^{\text{stance}} \in$

$\{\text{trusting}, \text{neutral}, \text{hostile}\}$. These states are not directly observed; the focal agent infers them from partner responses and payoff observations. Partners themselves are environment-side processes: they sample cooperate-or-defect responses from type- and stance-conditioned probabilities and undergo investment-dependent stance transitions, but do not perform variational inference or maintain affective precision estimates themselves. The present study thus centers on the focal agent's precision mechanism, whereas future work may extend this implementation to reciprocal active-inference partners.

With endowment $E = 10$ and return multiplier $M = 3$, when the partner cooperates, the invested amount is multiplied by M and split evenly, so the focal agent receives $Ma/2$; when the partner defects, the focal agent loses the invested amount. Focal payoff is

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases}$$

Thus, cooperation yields $\pi(a, \text{cooperate}) = 10 + 0.5a$, whereas defection yields $\pi(a, \text{defect}) = 10 - a$. The no-investment action $a = 0$ gives a risk-free payoff of 10, while higher investment increases both potential gains and potential loss.

Investment also affects future partner stance. Higher investment provides stronger cooperative evidence and shifts stance dynamics toward trust-building transitions; lower investment or withholding shifts dynamics toward trust-damaging transitions. Formally, the transition model interpolates between high-investment and low-investment endpoint matrices according to $\rho(a) = a/5$, with environment-side transition matrices specified in Appendix A.

2.2 Per-partner generative models

Rather than maintaining a single pooled social model, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k. This factorization keeps observations, beliefs, and confidence evidence local to the relationship that generated them. Each $\mathcal{M}_k$ uses the standard $A/B/C/D/E$ components of discrete active inference [6]: observation likelihoods, transition dynamics, preferences, initial priors, and policy priors. The focal agent's partner-local POMDP specification is provided in Appendix B.

Each partner model contains partner type, partner stance, and an own-investment state factor. Type and stance are inferred from evidence; own investment is deterministically set by the agent's executed action and treated as fully observed. This factor lets the payoff likelihood condition on the investment level the agent just chose, so the agent is uncertain about partner type and stance but not about its own investment. Observations are the partner's cooperate-or-defect response and the focal agent's realized payoff. The partner-response likelihood depends on type and stance, so predictive reliability is defined by response model fit rather than payoff: a reliably cooperative and a reliably defecting partner can both be predictable. The payoff likelihood depends on investment level and marginalizes over partner response. The agent's transition model mirrors the process dynamics in Appendix A, while scheduled betrayal and repair manipulations are environment-side interventions defined in Appendix D. This model supplies ordinary partner-state inference; the affective-precision layer in Section 2.3 estimates how confidently each partner model should be deployed during policy selection.

2.3 Partner-local affective precision

Partner-local affective precision maps relationship-specific model fit onto policy precision. At each round t, before assimilating the observed partner response, the focal agent evaluates how surprising that response would be under its predictive type–stance posterior for partner k. After this precision update, the partner-state posterior is updated on the same observation. Unlike prior affective-precision accounts centered on policy-inference quantities [13], the evidence source here is perceptual partner-response model fit. Because this update is computed separately for each partner, confidence can diverge across relationships even when the focal agent's task and preferences remain fixed.

Let $\hat{o}^{\text{act}}_{k,t}$ denote the observed response from partner k at round t and $q^{-}_{k,t}$ the type–stance posterior before updating on $\hat{o}^{\text{act}}_{k,t}$. The agent computes partner-response likelihood surprisal as the negative log posterior predictive probability assigned to that response:

$$\epsilon_{k,t} = -\log \sum_{s^{\text{type}},s^{\text{stance}}} P\big(\hat{o}^{\text{act}}_{k,t} \mid s^{\text{type}}, s^{\text{stance}}\big)\, q^{-}_{k,t}(s^{\text{type}}, s^{\text{stance}}). \tag{1}$$

Equation (1) is the negative log posterior predictive likelihood assigned to the observed partner response under $q^{-}_{k,t}$, marginalizing over the agent's pre-update beliefs about partner type and stance. Partner response directly tests the type-by-stance model: a reliably defecting partner can produce low surprisal despite poor payoff, while an erratic cooperative partner can produce high surprisal despite favorable outcomes. The signal therefore tracks partner-response model fit rather than partner value.

Per-round update order is: (i) plan from $q^{-}_{k,t}$; (ii) observe $\hat{o}^{\text{act}}_{k,t}$; (iii) compute $\epsilon_{k,t}$ and update β_k; (iv) assimilate $\hat{o}^{\text{act}}_{k,t}$ into the partner-state posterior.

The neutral baseline is the squared surprisal of a chance-level two-outcome partner-response prediction, $\sigma_0^2 = (-\log 0.5)^2$. Because the response modality is binary, probability 0.5 corresponds to predicting neither cooperation nor defection better than chance. Squaring places surprisal on a nonnegative error scale before signed affective charge is computed:

$$\phi_{k,t} = \alpha\big(\sigma_0^2 - \epsilon_{k,t}^2\big)\,, \tag{2}$$

where $\alpha \geq 0$ is a gain parameter. Positive $\phi_{k,t}$ means partner k's response was predicted better than chance; negative $\phi_{k,t}$ means it was predicted worse than chance.

The signed charge $\phi_{k,t}$ is then used as evidence over the partner-specific inverse-precision estimate β_k. Positive charge shifts posterior mass toward lower β_k, corresponding to higher policy precision; negative charge shifts posterior mass toward higher β_k, corresponding to lower policy precision. A persistence prior smooths this update across rounds so that confidence changes gradually rather than resetting after each interaction. The full categorical update is given in Appendix C.

The agent then sets partner-specific policy precision from the posterior mean:

$$\begin{aligned} \bar{\beta}_k &= \sum_{\ell} \beta_\ell \, q(\beta_\ell), \\ \gamma_k &= \gamma_{\text{base}} / \bar{\beta}_k, \end{aligned} \tag{3}$$

where γ_{base} is the baseline policy precision and $q(\beta_\ell)$ is the categorical mass on inverse-precision level β_ℓ. Because β_k is inverse precision, higher $\bar{\beta}_k$ yields more tentative policy commitment, whereas lower $\bar{\beta}_k$ yields higher γ_k and sharper policy commitment.

The β_k posterior is maintained as an auxiliary precision tracker rather than as a hidden state inside the focal agent's generative model. This keeps the mechanism lightweight and experimentally isolable, but prevents the agent from forming prior expectations over its own future confidence states; this limitation is discussed in Section 4.

2.4 Cross-partner policy selection

In partner-choice regimes, the agent must decide which partner-specific policy candidate to execute. Each per-partner model $\mathcal{M}_k$ first evaluates policies $\pi \in \Pi_k$ directed toward partner k, producing an unnormalized policy-evidence score $u_{k,\pi}$ for each candidate. The cross-partner step then combines these partner-indexed candidates into a single choice set. This step does not add a new state-inference model; it specifies how partner-local precision γ_k enters the final comparison among partner–investment candidates.

To apply γ_k locally, we separate each partner's mean policy evidence from the relative differences among that partner's policies. Let $\bar{u}_k = \frac{1}{|\Pi_k|} \sum_{\pi \in \Pi_k} u_{k,\pi}$ be the mean policy evidence for partner k. The precision-adjusted score is then

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k \right), \tag{4}$$

where γ_k is the partner-specific policy precision from Eq. (3). Partner-specific precision γ_k sharpens or flattens commitment among policies involving partner k, without by itself changing the average evidence for selecting that partner. Higher γ_k amplifies within-partner differences, producing more decisive policy commitment; lower γ_k compresses them, producing more tentative commitment.

The final policy posterior is computed by applying a softmax over the transformed scores $\tilde{u}_{k,\pi}$ across all partners and candidate policies. This procedure does not change the per-partner generative models; it only determines how partner-local precision enters cross-partner policy comparison. Reported policy entropy is the natural-log Shannon entropy of $q(\pi)$ over all partner–policy candidates at planning horizon H=4, averaged over rounds and seeds; values are in nats and are not normalized by maximum entropy.

2.5 Simulation setup

Simulations compare four precision-runtime variants: full partner-local precision, a shared-β ablation that pools evidence across partners while keeping per-partner POMDP beliefs intact, a tracked-only ablation that updates $q(\beta_k)$ but fixes $\gamma_k = \gamma_{\text{base}}$, and a no-affect control that disables the β tracker entirely. Conditions include open graded investment, partner choice, abrupt betrayal, and precision-dynamics/profile diagnostics. Full condition definitions, episode lengths, and metrics are provided in Appendix D. Code and experiment configurations are provided in the anonymized supplementary repository.

3 Results

Results test whether partner-local precision tracks predictability over payoff, acts through the $\beta_k \to \gamma_k$ deployment pathway, and remains behaviorally consequential under partner choice, betrayal, and profile variation.

3.1 Partner-local precision tracks predictability rather than realized payoff

Partner-local precision was more strongly associated with partner-response likelihood surprisal than with realized payoff. Because partner-choice episodes couple payoff, surprisal, and exposure, we computed active-encounter partial correlations controlling for encounter count. The precision–surprisal association remained strong ($r = -0.940$), while the precision–payoff association was near zero ($r = 0.023$). Negative signs indicate that higher partner-response likelihood surprisal corresponds to lower β_k-derived policy precision; Figure 1 plots absolute magnitudes for readability.

Pooling precision evidence across partners weakened this relationship-specific signal. In the shared-β ablation, per-partner beliefs remained separate but the precision tracker was updated from pooled partner evidence. The precision–surprisal association fell to $r = -0.496$ after controlling for encounter count, while the precision–payoff association was comparable in magnitude ($r = 0.535$). Thus, the update rule alone is not sufficient: when precision evidence is pooled, β_k no longer cleanly indexes relationship-local model fit and becomes more entangled with global exposure/payoff structure.

Payoff in the same predictability-value analysis run did not identify the mechanism. Although both precision variants had higher analysis-window payoff than no-affect in this run, payoff did not isolate the mechanism: partner-local and shared-β were nearly tied in payoff while differing sharply in their precision–surprisal signatures. The payoff panel therefore complements the correlation panels: payoff can vary across runtime variants, but the clearest relationship-specific precision signal is the partner-local precision–surprisal association, not a precision–payoff association. This is consistent with affective precision serving as a confidence-calibration signal rather than a direct reward-maximization signal.

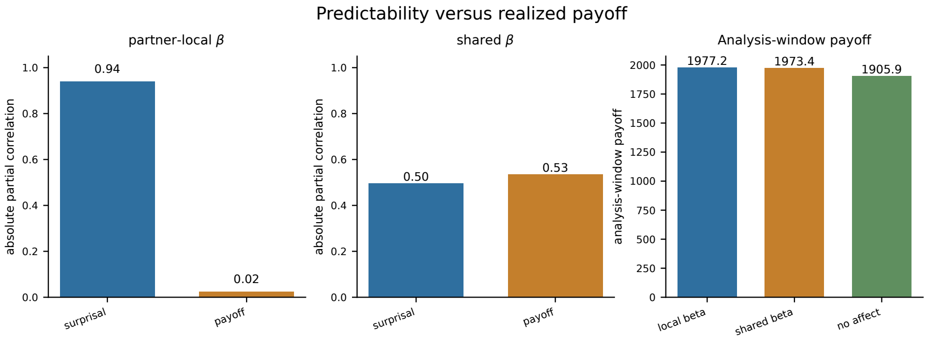

Fig. 1: Partner-local precision is more strongly associated with partner-response likelihood surprisal than with realized payoff. The correlation panels plot absolute active-encounter partial correlations; the payoff panel shows payoff in the same predictability-value analysis run. Pooling precision evidence across partners attenuates the partner-local predictability signature and makes the pooled precision signal less diagnostically specific, with payoff and surprisal associations becoming comparable under shared β.

3.2 Tracked-only ablation isolates policy commitment

To test whether partner-local affective precision changes behavior through policy deployment rather than partner-state inference, we used a tracked-only ablation. This variant updates $q(\beta_k)$ normally while holding $\gamma_k = \gamma_{\text{base}}$, preserving the precision estimate but preventing it from modulating policy selection.

The tracked-only ablation confirmed that the tracker can move without producing the deployment effect. In the open graded deployment run, tracked-only produced a comparable average within-episode max–min range of posterior mean $\bar{\beta}_k$ (1.34 in both variants). The key difference was deployment: full precision modulation lowered policy entropy by 0.238 relative to tracked-only, while cumulative payoff remained nearly matched (1868.3 versus 1866.2). Thus, β_k matters behaviorally when it is allowed to modulate γ_k, producing sharper policy commitment. When that pathway is cut, the tracker remains active but the entropy effect is lost. Figure 2 shows the pathway-isolation comparison.

This supports the interpretation of affective precision as a calibration layer rather than an inference-improvement mechanism. The tracker does not directly improve partner-state inference; it regulates how strongly the agent commits to policies based on its current partner model. Whether sharper commitment improves payoff depends on the social context, not on precision alone.

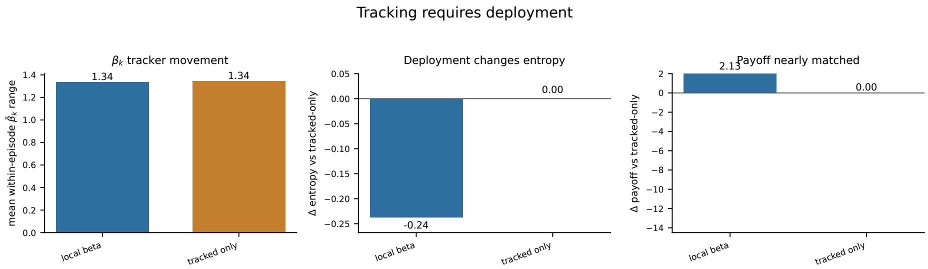

Fig. 2: Open graded regime: tracked-only and partner-local precision show a comparable average within-episode max–min range of posterior mean $\bar{\beta}_k$, but only partner-local precision deploys that tracker through γ_k. Relative to tracked-only, partner-local precision lowers policy entropy without improving payoff, isolating the entropy effect to the $\beta_k \rightarrow \gamma_k$ policy-precision pathway.

3.3 Partner choice extends policy commitment to social allocation

Partner-local precision also affected policy commitment when the agent could choose whom to engage. In the partner-choice regime, each candidate pairs a partner with a multi-step investment plan, of which only the first step is executed each round. Precision modulation lowered policy entropy relative to the no-affect control (8.60 versus 8.83), indicating that the $\beta_k \rightarrow \gamma_k$ pathway shapes partner–investment policies, not only investment levels within a fixed interaction.

The allocation shift was small and condition-specific rather than a simple cooperator-seeking effect. Across selected partner types, partner-local precision produced a more balanced allocation than the no-affect control (cooperator 25.3% versus 29.2%; exploiter 25.5% versus 21.3%; reciprocator 23.7% versus 22.5%; random 25.5% versus 27.1%). This should not be read as a general preference for any one partner type. The main effect is sharper commitment under current partner models, not direct value tracking; aggregate payoff was nearly unchanged (1868.3 versus 1866.2).

Together, these results show that partner-local precision can reshape engagement without guaranteeing reward improvement. Whether sharper commitment helps or hurts depends on whether the social environment remains stable. The abrupt-betrayal condition tests this boundary directly.

3.4 Abrupt betrayal tests confidence under change

Accumulated confidence helps when it remains attached to a valid partner model, but can lag behind abrupt social change. The abrupt-betrayal condition tests this boundary: a previously cooperative partner switches stance at round 31, and the agent must adapt while carrying confidence built from the pre-switch relationship.

Precision modulation continued to affect deployment after the switch. Mean policy entropy was lower under partner-local precision than under the no-affect control (8.36 versus 8.74), with a bootstrap interval that did not cross zero (-0.63 to -0.16). Joint partner-type accuracy was also higher under precision modulation (0.372 versus 0.266; bootstrap interval 0.024 to 0.188), consistent with engagement being redirected rather than simply disrupted. Because the tracked-only ablation in Section 3.2 isolated the $\beta_k \to \gamma_k$ pathway, this accuracy difference is best interpreted as a downstream effect of changed engagement and sampling, not as direct improvement to the partner-state update.

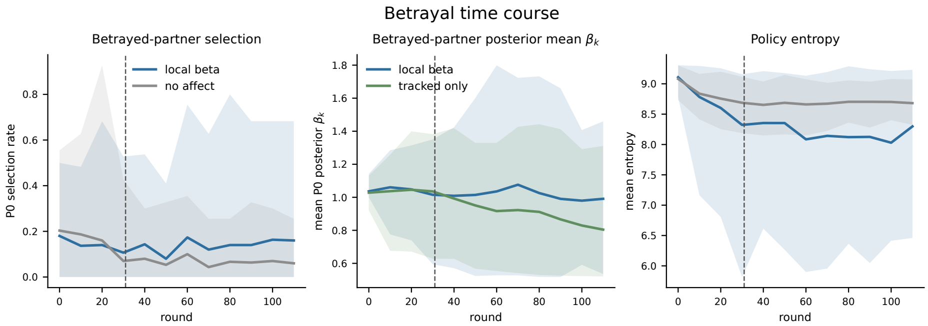

Fig. 3: Abrupt betrayal in the partner-choice regime. The dashed line marks the round-31 stance switch for partner 0. Shaded bands show 95% intervals across seeds. Partner-local precision maintains lower policy entropy after the switch, while the tracked-only condition shows that β_k dynamics can persist without the same deployment effect. Betrayed-partner selection illustrates the reallocation problem: confidence remains behaviorally relevant after the relationship changes, but payoff depends on how engagement is redistributed.

Payoff showed a weaker and more variable pattern. The payoff difference was positive but uncertain (1185.9 versus 1172.1; bootstrap interval -21.6 to 52.0). This pattern is consistent with a calibration mechanism: confidence can keep policies decisive after a relationship changes, but reward depends on whether the agent redirects engagement quickly enough and toward useful alternatives (Figure 3).

The betrayal result exposes a timing problem in social confidence: confidence built from reliable past evidence can remain behaviorally active after the relationship changes. We next vary precision gain and prior model fitness to examine how confidence-revision dynamics differ across agents.

3.5 Precision dynamics as computational profiles of social trust calibration

The betrayal result shows that confidence revision has temporal structure: confidence can be too weak, too slow, or too reactive. Varying precision gain α confirmed that gain controls the amplitude of β_k dynamics: in the betrayal regime, the average within-episode max–min range of posterior mean $\bar{\beta}_k$ increased from 0.147 at $\alpha = 0.05$ to 1.218 at $\alpha = 8.0$. Payoff did not follow this monotonic pattern, indicating that faster confidence revision is not automatically better revision.

Crossing prior model fitness with gain produced distinct trust-calibration profiles. Low gain muted confidence dynamics and weakened policy commitment; high gain amplified revision but increased vulnerability to overcommitment after change; cautious priors suppressed premature confidence but also reduced exploitation of reliable partners. In the tested betrayal profiles, the default configuration achieved the highest payoff, while naive-stubborn, avoidant-rigid, and anxious-reactive variants expressed different failure modes: diffuse commitment, suppressed exploitation, and post-change overcommitment. Exact priors, gain values, and profile metrics are reported in Appendix E.

Trust-repair diagnostics further separated reengagement from restored confidence. Some variants reengaged a reverted partner without rebuilding the same confidence dynamics, while high-gain variants showed larger β_k swings without faster payoff recovery. The agent can return to a partner before it trusts them again. Extended profile and repair metrics are reported in Appendix E.

4 Discussion

The results support partner-local affective precision as a calibration mechanism for social policy commitment. Precision was defined from partner-response predictability rather than payoff, lost its entropy effect when the $\beta_k \to \gamma_k$ pathway was cut, and remained behaviorally active under abrupt change without guaranteeing reward improvement. Together, these findings support the paper's central distinction: social prediction and social commitment are related but distinct. A reliably cooperative partner and a reliably defecting partner can both generate low partner-response likelihood surprisal because both are predictable; conversely, a favorable but inconsistent partner can remain difficult to model. Partner-local affective precision therefore estimates how well the current partner model is working, then regulates how strongly that model is expressed in policy.

The tracked-only ablation clarifies where the mechanism acts. Updating β_k without allowing it to modulate γ_k preserved the tracker but removed the entropy effect, while full precision modulation sharpened policy commitment. This supports the interpretation that partner-local affective precision does not change the partner-state update rule directly; any accuracy differences that emerge under partner choice are downstream sampling effects, reflecting which partners the agent engages and observes, while the mechanism itself changes how strongly current partner beliefs are deployed. It also clarifies the distinction from affective-precision accounts centered on policy-inference quantities: here, the model-fitness signal is partner-response likelihood surprisal under the pre-update predictive type–stance posterior $q_{k,t}^{-}$.

The abrupt-betrayal condition shows the main limitation of this first calibration layer. Confidence built from reliable past evidence can remain behaviorally active after the relationship changes. In the present model, β_k is an auxiliary tracker, so the agent does not explicitly infer when its own confidence has become stale. A fuller architecture should add volatility or change detection that discounts accumulated confidence when prediction-error structure shifts, and should make partner-choice policies more exploratory after detected change. Embedding β_k as a hidden state would also allow the agent to form prior expectations over its own future confidence rather than only updating confidence retrospectively.

The profile analyses generate computational hypotheses about trust calibration, with potential links to work on social learning, psychosis, and attachment [3, 1, 5]. Precision gain and prior model fitness shaped update amplitude, sensitivity to change, and reengagement after repair, but no single profile was uniformly best. High gain expanded confidence swings without monotonic payoff improvement; low gain stabilized behavior but slowed revision; cautious priors protected against premature confidence while suppressing confidence even toward reliable partners. These tradeoffs suggest empirical tests on trust-game time series, including repair settings after trust violations: whether human behavior separates predictability from value, policy deployment from partner-state inference, and reengagement from restored confidence. A further extension is to replace scripted partners with reciprocal active-inference agents that maintain their own beliefs, policies, and precision estimates.

5 Conclusion

We formalize social affect as a relationship-specific metacognitive signal: an estimate of how confidently an agent should select policies from its current model of each partner. Implemented as partner-local affective precision, this signal updates from partner-response likelihood surprisal and modulates policy precision during selection. Across simulations in a graded multi-partner trust game, precision tracks predictability more closely than payoff, affects behavior through the $\beta_k \to \gamma_k$ policy-precision pathway rather than improved partner-state inference, and remains behaviorally active when relationships change.

These results support a distinction between social prediction and social commitment. An agent can infer what a partner is likely to do while separately estimating how strongly that inference should guide policy commitment. Precision gain and prior model fitness shape this calibration process, producing tradeoffs in confidence accumulation, betrayal adaptation, reengagement, and repair. Future empirical work can test whether human trust-game behavior shows similar distinctions between predictability and value, inference and policy commitment, and reengagement and restored confidence.

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Appendix

Simulation code, experiment configurations, analysis scripts, figure-generation code, and row-level result files are available in the anonymous supplementary repository.

A Trust-Game Generative Process

This appendix specifies the environment-side trust-game process: scripted partners with hidden type and stance, investment-dependent stance transitions, binary responses, and deterministic payoff delivery. Appendix B specifies the focal agent's POMDP model.

A.1 Episode structure

Each episode consists of repeated interactions between the focal agent and four environment-side partners. On each round, the partner is assigned or selected, the focal agent chooses $a \in \{0, 1, 2, 3, 4, 5\}$, the partner emits $o_k^{act} \in \{\text{cooperate}, \text{defect}\}$, payoff is delivered by Eq. (5), and stance updates by Eq. (6).

A.2 Partner state variables

Each environment-side partner k has two process variables hidden from the focal agent: behavioral type $s_k^{\text{type}} \in \{\text{cooperator}, \text{reciprocator}, \text{exploiter}, \text{random}\}$ and stance toward the focal agent $s_k^{\text{stance}} \in \{\text{trusting}, \text{neutral}, \text{hostile}\}$. These govern how the environment samples partner responses and updates stance over the episode; type sets the partner's broad response tendency, and stance modulates how cooperative those responses are toward the focal agent at the current interaction.

A.3 Partner-response probabilities

The partner cooperation likelihood table defines the probability the partner cooperates given type and stance:

Table 1: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

The table defines the environment-side cooperation probability $P(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}})$, with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The same response probabilities are used in the focal agent's partner-response likelihood A_k^{act} (Appendix B.3), allowing the focal agent to infer type and stance from observed cooperate-or-defect responses.

A.4 Payoff function

Payoff is delivered by the environment after the partner response is sampled. With endowment $E = 10$ and multiplier $M = 3$, the invested amount is subtracted from the focal agent's endowment. If the partner cooperates, the investment is multiplied by M and split evenly, so the focal agent receives $Ma/2$. If the partner defects, the focal agent receives no return from the investment.

$$\pi(a, o_k^{\text{act}}) = \begin{cases} E - a + \frac{Ma}{2}, & o_k^{\text{act}} = \text{cooperate}, \\ E - a, & o_k^{\text{act}} = \text{defect}. \end{cases} \tag{5}$$

Thus, $a = 0$ is risk-free, while higher investment increases both possible gains and potential loss.

A.5 Stance transition dynamics

Stance transitions depend on the focal agent's graded investment a by interpolating between a high-investment/trust-building endpoint and a low-investment/withholding endpoint:

$$\rho(a) = \frac{a}{|\mathcal{A}| - 1} = \frac{a}{5}, \qquad B^{\text{stance}}(a) = \rho(a)B^{\text{high}} + (1 - \rho(a))B^{\text{low}}. \tag{6}$$

High investment shifts dynamics toward trust building; low investment or withholding shifts dynamics toward trust damage; recovery from hostility is slower than collapse.

Table 2: Endpoint stance transition dynamics. The focal agent's graded investment interpolates between B^{high} and B^{low} according to Eq. (6).

High investment / trust-building endpoint B^{high}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Low investment / withholding endpoint B^{low}			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: sustained high investment gradually builds trust, while sustained withholding rapidly damages trust; recovery from hostility is slow. In scheduled betrayal and repair protocols, the environment can additionally intervene on a partner's stance according to the protocol schedule; these interventions are not known transition events in the focal agent's POMDP.

A.6 Boundary of the current simulation

All reported simulations use one focal active-inference agent interacting with four scripted partners that maintain hidden type and stance, sample responses by Appendix A.3, and update stance by Appendix A.5; they do not infer policies or maintain affective precision.

B Focal-Agent POMDP Specification

This appendix specifies the focal agent's partner-local generative model in standard discrete active-inference form. Each partner k has a POMDP $\mathcal{M}_k$ with hidden type, stance, and own-investment state factors. The model follows Appendix A but is not identical to the environment: payoff is an observation modality marginalizing over partner response; scheduled betrayal and repair are environment-side; affective precision is external to the POMDP.

B.1 Partner-local factorization

The focal agent maintains a separate model M_k for each partner, with partner-specific likelihoods, transitions, preferences, priors, and policy priors. Partner choice is handled outside the individual POMDPs by comparing partner–investment policies after each M_k has produced policy evidence.

B.2 Hidden state factors and observations

As described in Section 2.2, the focal agent's generative model for each partner k maintains three hidden factors:

$$s_k^{\text{type}} \in \mathcal{S}^{\text{type}}, \tag{7}$$
$$s_k^{\text{stance}} \in \mathcal{S}^{\text{stance}}, \tag{8}$$
$$s^{\text{own}} \in \mathcal{A},$$

where

$$\begin{aligned} \mathcal{S}^{\text{type}} &= \{\text{cooperator}, \text{reciprocator}, \text{exploiter}, \text{random}\}, \\ \mathcal{S}^{\text{stance}} &= \{\text{trusting}, \text{neutral}, \text{hostile}\}, \\ \mathcal{A} &= \{0, 1, 2, 3, 4, 5\}. \end{aligned}$$

s^{own} indexes executed investment, is deterministically set by the agent's action, and is treated as fully observed. The joint hidden state per partner spans $4 \times 3 \times 6 = 72$ configurations.

Observations are partner response and focal payoff:

$$o_k^{\text{act}} \in \mathcal{O}^{\text{act}},$$
$$o_k^{\text{pay}} \in \mathcal{O}^{\text{pay}},$$

where $\mathcal{O}^{\text{act}} = \{\text{cooperate, defect}\}$ and $\mathcal{O}^{\text{pay}}$ comprises eleven discrete payoff levels (Eq. (5)).

B.3 Observation likelihoods

Partner-response likelihood A_k^{act}. A_k^{act} uses the cooperation probabilities in Table 1 and is uniform over s^{own}; investment affects future responses indirectly through stance transitions.

Payoff likelihood A_k^{pay}. In the environment, payoff is deterministic given investment and partner response (Appendix A.4). In the focal agent's POMDP, payoff is represented as a likelihood over payoff observations conditional on own investment and latent partner state, marginalizing over partner response:

$$\begin{aligned} &P(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}) \\ &\quad = \sum_{o^{\text{act}} \in \{\text{cooperate,defect}\}} \mathbb{I}\big[\pi(s^{\text{own}}, o^{\text{act}}) = v\big]\, P(o_k^{\text{act}} = o^{\text{act}} \mid s_k^{\text{type}}, s_k^{\text{stance}}) \end{aligned} \tag{9}$$

This makes higher investment beneficial when cooperation is likely and costly when defection is likely, while keeping payoff preferences inside the generative model.

B.4 Transition dynamics

Type drift B_k^{type}. Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{10}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal scenarios set $p_{\text{switch}} = 0$.

Stance dynamics B_k^{stance}. $B_k^{\text{stance}}(a)$ mirrors the interpolation in Eq. (6), using the endpoint matrices in Table 2. Scheduled betrayal and repair intervene on the environment-side process only; the focal agent is not given the intervention times.

Own-investment state factor B^{own}. B^{own} deterministically sets own investment to the executed action:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{11}$$

B.5 Preferences, priors, and policy priors

Observation preferences C_k. There is no direct preference over partner responses:

$$C_k^{\text{act}} = [0,\, 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{12}$$

Payoff preferences ascend over the graded payoff support generated by Eq. (5):

$$C_k^{\text{pay}} = \log \text{softmax}(\mathbf{u}/\tau)\,, \quad \mathbf{u} = (5, 6, 7, 8, 9, 10, 10.5, 11, 11.5, 12, 12.5), \tag{13}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct preferences over partner responses.

Initial-state priors D_k. Type, stance, and own-investment priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{14}$$
$$D_k^{\text{stance}} = [0.2,\, 0.6,\, 0.2], \tag{15}$$
$$D^{\text{own}} = \text{Unif}(\{0, 1, 2, 3, 4, 5\}). \tag{16}$$

Policy priors E_k. Policy priors are uniform over admissible graded investment sequences at the configured planning horizon. In partner-choice settings, partner selection is handled outside the per-partner POMDP by evaluating partner–investment policy candidates with precision-adjusted scores before executing the chosen investment. This cross-partner comparison is specified in Appendix C.

B.6 Runtime and precision boundary

The partner-local POMDPs supply ordinary active-inference belief updating and policy evidence for each partner. They do not contain β_k as a hidden state. Instead, β_k is maintained as an auxiliary partner-local precision tracker, updated from partner-response likelihood surprisal and then mapped to policy precision γ_k. This design keeps the POMDP state space compact and makes the precision pathway experimentally isolable. The cost is that the focal agent cannot form prior expectations over its own future confidence states. Appendix C specifies the auxiliary update and the cross-partner policy-selection procedure.

C Affective-Precision Update and Policy Selection

This appendix specifies the auxiliary affective-precision update layered on top of the focal agent's partner-local POMDPs. The update maps partner-response model fit into a categorical posterior over inverse precision β_k, then maps the posterior mean to partner-specific policy precision γ_k. The same appendix also specifies how γ_k enters partner-choice policy comparison. At each round t, affective charge is computed from Eq. (1) using the pre-update predictive posterior $q^{-}_{k,t}$ before the partner-state posterior is updated on $\hat{o}^{\mathrm{act}}_{k,t}$ (Section 2.3).

C.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete inverse-precision levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The categorical representation keeps the inverse-precision convention explicit, bounds the precision dynamics, and allows the update to be analyzed as movement over interpretable confidence states.

C.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\mathrm{prev}}(\beta_k), \tag{17}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

C.3 Charge-based pseudo-likelihood

Second, a charge-based pseudo-likelihood maps positive charge toward high policy precision (low β) and negative charge toward low policy precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{18}$$

where β_ℓ ranges over the five support points. This auxiliary update satisfies the intended monotonicity: positive charge reinforces lower β, negative charge reinforces higher β, and the effect scales with effective precision at each support point.

C.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \tag{19}$$

Policy precision is then set from the posterior mean as given in Eq. (3) of the main text. The quantity β_k is inverse precision, so higher posterior mean β_k lowers policy precision γ_k, while lower posterior mean β_k raises γ_k. As noted in Appendix B.6, β_k is maintained as an auxiliary tracker outside the POMDP.

C.5 Cross-partner policy selection

In partner-choice regimes, partner-specific precision sharpens or flattens policy commitment among policies directed toward a given partner while preserving the mean policy evidence associated with that partner. The transformation in Eq. (4) applies γ_k to deviations from the partner-specific policy mean, rather than to raw policy scores. The agent samples or selects policies from the combined set of transformed scores across all partners and policies. This lets γ_k control commitment among policies involving partner k without changing the average evidence for approaching partner k during cross-partner comparison.

C.6 Ablation variants

Runtime variants are summarized in Table 3 (see also Section 2.5).

Table 3: Precision-runtime ablation variants.

Variant	β tracker and γ_k	Precision scope
No-affect	off; $\gamma_k = \gamma_{\text{base}}$ fixed	—
Tracked-only	$q(\beta_k)$ updated; $\gamma_k = \gamma_{\text{base}}$ fixed	per-partner
Partner-local	$q(\beta_k)$ updated; γ_k from $q(\beta_k)$	per-partner
Shared-β	pooled $q(\beta)$ updated; γ_k from shared β	pooled tracker; separate beliefs

D Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

D.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, graded investment actions, and environment-side partner policies (Appendices A–B). Conditions differ by assignment mode, scheduled partner changes, and precision/profile manipulations within the graded payoff regime. In conditions without explicit partner choice, the interaction partner is assigned according to the configured protocol; in partner-choice conditions, the agent selects the partner before selecting an investment level. Table 4 summarizes the condition families.

Table 4: Simulation condition families and purpose.

Condition	Purpose
Open graded	Deployment contrast among affect, no-affect, and tracked-only
Locality probe	Precision–surprisal vs. precision–payoff correlations under local/shared β
Partner choice	Social allocation and partner–investment policy entropy
Abrupt betrayal	Post-switch entropy, payoff, reallocation, and type accuracy
Profile variation	α/prior manipulations for trust-calibration profiles
Forgiveness	Reengagement and β recovery after repair

D.2 Seed counts and episode lengths

Behavioral comparisons use 30 seeds per condition and profile sweeps use 20 seeds (Table 5).

Table 5: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
Open deployment	graded, random/open choice	200	30
Locality probe	graded partner choice	200	30
Partner allocation	partner choice	200	30
Abrupt betrayal	scheduled partner-choice betrayal	120	30
Precision perturbation	graded perturbations	200	30
Profile program	sweeps, factorial, forgiveness	200	20

D.3 Betrayal schedule

In the abrupt-betrayal protocol, P0 begins as a trusting cooperator and switches to hostile at round 31; other partners retain their initial configurations and $p_{\mathrm{switch}} = 0$. Primary readouts are post-switch payoff, entropy, reallocation, and joint partner-type accuracy. The forgiveness protocol extends this schedule with baseline rounds 1–80, betrayal rounds 81–120, and repair/reversion rounds 121–200.

D.4 Metrics

Table 6 defines reported metrics.

Table 6: Metric glossary.

Metric	Definition
Total payoff	Cumulative focal payoff over episode/window
Policy entropy	Natural-log Shannon entropy of $q(\pi)$, averaged over rounds/seeds
Joint partner-type accuracy	Mean correctness of inferred partner type
Selection share	Fraction of rounds each partner is chosen
Selection concentration	Gini coefficient over partner-selection distribution
Mean within-episode $\bar{\beta}_k$ range	Average within-episode max–min range of posterior mean $\bar{\beta}_k$, averaged across seeds and episodes
Investment churn	Fraction of adjacent rounds with changed investment
Early exploiter investment	Rounds 1–30 choosing exploiter with investment $\geq$ half range
Trust asymmetry	Withdrawal latency / initial high-investment approach latency
Recovery rounds	Rounds after switch until selection/payoff returns to threshold
Locality-probe partial (r)	Partner–seed correlation between β-derived precision and surprisal/payoff, controlling for encounter count

E Extended Profile and Robustness Results

This appendix reports extended sweeps, profiles, and robustness diagnostics.

E.1 Precision-dynamics parameter variation

These auxiliary controls vary the amplitude and stability of β_k dynamics.

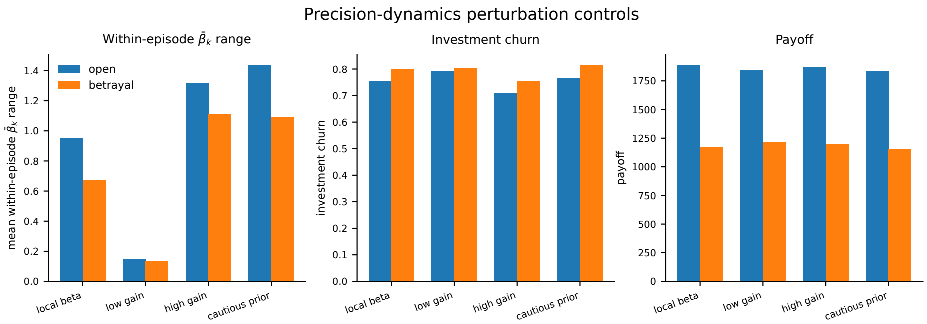

Fig. 4: Perturbation controls in the graded decision regime. Low gain compresses the average within-episode max–min range of posterior mean $\bar{\beta}_k$, while a wider average within-episode max–min range of posterior mean $\bar{\beta}_k$ does not map monotonically to payoff.

Table 7: Precision-dynamics parameter variation in the graded decision regime.

Variant	Mean within-episode $\bar{\beta}_k$ range	Total payoff	Policy entropy
affect	0.95	1884.6	7.94
low-gain	0.15	1840.7	8.84
high-gain	1.32	1871.1	7.63
cautious-prior	1.44	1834.1	8.28

E.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. The average within-episode max–min range of posterior mean $\bar{\beta}_k$ increases monotonically with α in the betrayal regime:

Table 8: Mean within-episode $\bar{\beta}_k$ range by precision-gain α (betrayal regime).

α	Mean within-episode $\bar{\beta}_k$ range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

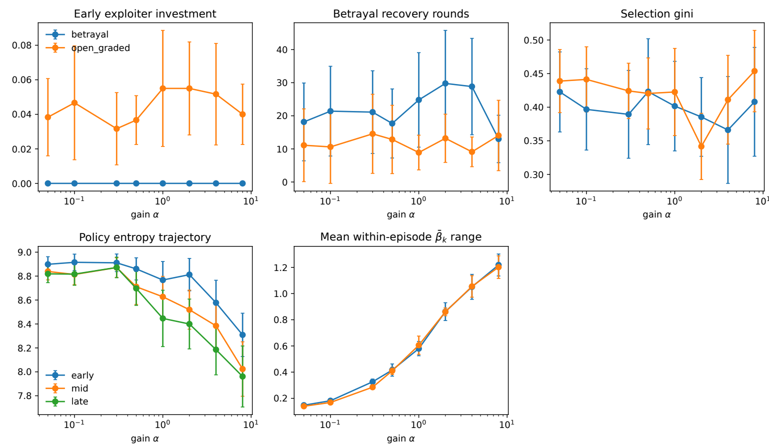

Fig. 5: Precision-gain sweep across open graded and betrayal environments. Higher α expands the average within-episode max–min range of posterior mean $\bar{\beta}_k$, while engagement, recovery, entropy, and payoff readouts vary non-monotonically. Precision gain therefore controls confidence-reactivity amplitude rather than mapping directly to reward.

E.3 Gain-prior profile signatures

"Naive" and "cautious" refer to priors over $q(\beta_k)$: naive $[0.40, 0.40, 0.15, 0.04, 0.01]$, cautious $[0.01, 0.04, 0.15, 0.40, 0.40]$. Low/high gain use $\alpha = 0.1/3.0$. Profile labels correspond to these combinations: naive-stubborn uses naive prior/low gain, anxious-reactive uses naive prior/high gain, avoidant-rigid uses cautious prior/low gain, and hypervigilant uses cautious prior/high gain. The default uses $\alpha = 3.0$ with centered initialization at $\beta_k = 1.0$. Table 9 reports raw metrics and Figure 6 shows column-normalized profile signatures.

Table 9: Behavioral metrics by profile (betrayal environment). Default agent shown for reference.

Profile	Payoff	Recovery rounds	Gini	Mean within-episode $\bar{\beta}_k$ range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

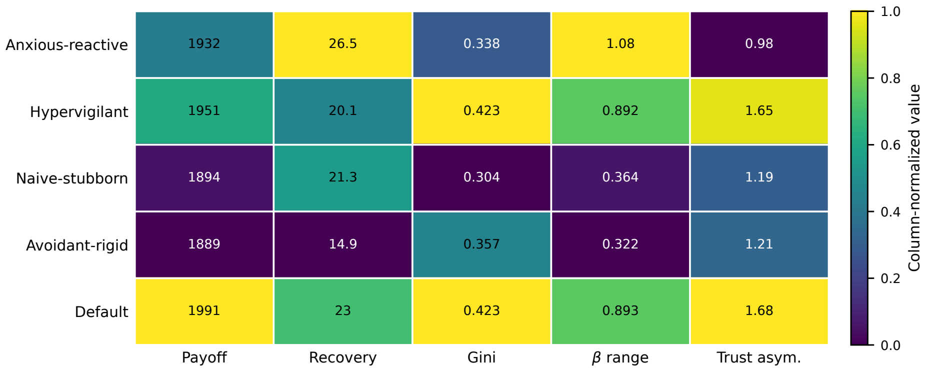

Fig. 6: Gain-prior profile signatures from the prior-by-α factorial (betrayal environment). Cells are column-normalized across profiles for visual comparison; raw metric values are reported in Table 9. The heatmap shows that gain and prior model fitness produce distinct calibration signatures rather than a single reward-aligned ordering.

E.4 Forgiveness and trust repair

The forgiveness protocol uses baseline rounds 1–80, betrayal rounds 81–120, and repair/reversion rounds 121–200. Table 10 reports reengagement, payoff recovery, latency, and mean within-episode $\bar{\beta}_k$ range; Figure 7 shows reengagement, β_k recovery, and payoff recovery.

Table 10: Forgiveness and trust-repair metrics by precision profile. Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	Mean within-episode $\bar{\beta}_k$ range
cautious-high-α	0.518	0.996	5.5	0.971
cautious-low-α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high-α	0.560	1.005	2.2	1.100
naive-low-α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	–

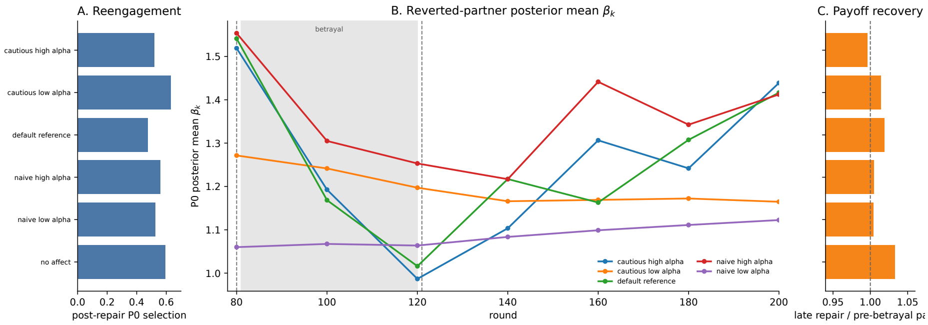

Fig. 7: Forgiveness/trust-repair diagnostic from the profile analyses. Vertical markers indicate the transition from cooperative baseline to betrayal and then to repair/reversion. Reengagement and payoff recovery are shown alongside the reverted-partner β_k trajectory. Because β_k is inverse precision, lower β_k indicates higher confidence / higher policy precision; no-affect is omitted from the β_k trajectory panel because it has no β tracker. The figure shows that agents can reengage a repaired partner before rebuilding the same confidence dynamics.